Project Title

Resume–Job Matching & Talent-Marketplace Engi

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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Submitted by G.Teja - 2520090162
P.Vedadri Naidu– 2520090060

Team Number: 5
Section: 11

Under the guidance of

Mrs. Ch. Anitha Asst. Professor, CS&IT



Hyderabad-500090, Telangana, India.

ABSTRACT

The **Resume–Job Matching & Talent-Marketplace Engine** is an intelligent platform for automating the recruitment process.It matches large numbers of résumés with suitable job postings efficiently. The system indexes and organizes candidate résumés and job requirements.It normalizes different skill names and vocabularies for accurate matching.Candidates are evaluated based on skills, qualifications, and experience.A compatibility score is calculated for each candidate-job combination.The system finds the optimal assignment of candidates to available roles.It also identifies the minimum skill set required to build a successful team.This reduces manual screening and improves recruitment efficiency. Overall, the system enables faster, smarter, and data-driven hiring decisions.By combining résumé indexing, skill normalization, intelligent matching, optimization, and skill-set analysis, the proposed engine provides a scalable and efficient solution for modern recruitment and talent management. The system aims to reduce manual screening efforts, improve hiring accuracy, and support data-driven recruitment decisions.

Signature of the Guide